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Editor-in-Chief *npj Precision Oncology*

Dear Editor,

We submit our manuscript "**An 8-gene RAI-responsiveness biomarker outperforms BRAF V600E status in papillary thyroid carcinoma**" for consideration as an Article. This is the v5 version, which adds (on top of the v4 BOOST sprint): (i) MSK-IMPACT 2017 cross-cohort TERT prevalence consistency check (n = 231 thyroid samples, six histologies); (ii) NCBI GEO candidate enumeration for revision-round meta-analysis extension (9 candidates passing n 30 + thyroid filter, including GSE310793 and the directly-relevant RAI-refractory-vs-avid GSE151180); and (iii) a Korean cohort prospect paragraph reflecting active outreach to Bundang SNUH and SNUH thyroid surgery / molecular pathology.

Decisions about radioactive-iodine (RAI) therapy in papillary thyroid carcinoma (PTC) currently rely on BRAF V600E mutation status and clinician-driven judgement about differentiation, with no quantitative pre-treatment biomarker that outperforms the mutation alone. We deliver an 8-gene panel of canonical thyroid-differentiation genes (NIS, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1) that achieves 5-fold cross-validated AUC 0.954 in DM1/DM2 cluster prediction (Random Forest 0.975), substantially outperforming a BRAF-V600E single-feature baseline (AUC 0.822, **ΔAUC = +0.132**) on 513 TCGA-THCA primary tumours.

Three reviewer-anticipated robustness layers added in v4 (Fig. 6E–G). 1. **Multi-cohort meta-analysis** across four independent transcriptomic cohorts (GSE27155 n = 99, GSE29265 n = 49, GSE33630 n = 105, GSE76039 n = 37; total n = 290). Random-effects logit-AUC pooling yields **pooled AUC = 0.980 (95% CI 0.869–0.997, I^2 = 0%, all 4 cohorts AUC > 0.96)**. The I^2 = 0% is a notably tight result for a thyroid expression biomarker — most prior transcriptomic panels show I^2 > 50%. 2. **Decision Curve Analysis** (Vickers & Elkin 2006) of 8-gene LogReg vs BRAF V600E vs treat-all/treat-none across decision thresholds 0.05–0.95 (n = 91). The 8-gene strategy is the **dominant strategy at all 91 thresholds**, the strongest possible Decision Curve outcome — translating the AUC outperformance into a clinical-utility statement. 3. **9-strata subgroup forest** across Age (<45, 45–65), Sex (Male, Female), Stage (I/II, III/IV), and Histology (cPTC, FVPTC) with bootstrap 95% CIs. **7 of 9 strata exceed AUC 0.85**; the highest-risk Stage III/IV subgroup peaks at **AUC 0.996** — exactly the subgroup pre-RAI decision-making is hardest in.

MSK-IMPACT cross-cohort prevalence consistency. We additionally queried the public MSK-IMPACT 2017 study (n = 231 thyroid samples) via the cBioPortal datahub and computed TERT promoter prevalence by histology: PTC 63.4%, PDTC 56.7%, ATC 81.8%, FTC 80%, HTC 56.5%, MTC 0%. The TCGA-THCA 7.1% PTC TERT figure (primary indolent disease) and the MSK-IMPACT 63.4% PTC TERT figure (advanced/recurrent/metastatic enrichment) are not in conflict — they bracket the natural-history range of TERT acquisition. ATC TERT% (81.8%) recapitulates the Landa 2016 finding. BRAF V600E in MSK-IMPACT PTC (64.5%) is consistent with TCGA's 56.1%. This cross-cohort context strengthens our R8 framing of $TERT^{+}$ as a Stage III/IV / advanced-disease molecular handle.

Korean cohort outreach in active progress. Bundang SNUH and SNUH thyroid surgery / molecular pathology are being engaged for ethnically-distinct cross-validation. Korean PTC BRAF prevalence (~62%) provides a third reference point distinct from both TCGA (56%) and MSK-IMPACT (64.5%). Wet-lab validation (qPCR/NanoString on FFPE archives) is in parallel outreach. We commit to incorporating Korean validation in the revision round upon data availability.

Browser-based interactive RAI calculator (`reports/v17_boost/rai_calculator.html`) — a self-contained client-side JavaScript implementation that embeds the trained LogReg coefficients and the TCGA-THCA reference distribution. Users enter $\log_2(\text{TPM} + 1)$ values for the eight genes and obtain a real-time logit P(DM2) score. Three preset profiles (TCGA median, DM1 prototype, DM2 prototype) allow non-computational reviewers to verify the calculator without input data. No server, no installation, no telemetry — intended both as a transparency artefact and as a usable clinical-translation tool.

External validation on GSE76039 (n = 37 PDTC + ATC) yields direction-correct AUC 0.974, confirming the panel's transferability. A composite hot/cold immune score (cytolytic activity + IFN- + immune fraction) cleanly separates DM1 (hot) from DM2 (cold) with **Cohen's d = +1.683 (Mann-Whitney p = 3.99×10 ¹)**.

Liu–Xing four-genotype integration with honest robustness audit. Beyond the DM1/DM2 axis, we recover 36 TERT-promoter-mutated patients via the cBioPortal `thca_tcga_pub` mirror (Sanger-validated by TCGA Cell 2014). A four-group stratification (BRAF / RAS / TERT / triple-negative) yields multivariate logrank p = 3.78 × 10 for overall survival; 1,000-iteration bootstrap median p = 2.6 × 10 and leave-one-out worst-case p < 10 ³ across all 36 TERT patients confirm the separation is not single-event-driven. We honestly report that the univariate TERT signal (Cox HR = 6.31, p = 3 × 10) drops to multivariate HR = 1.88 (p = 0.29) after stage + age + sex adjustment, and reframe TERT as a clinically actionable molecular handle for Stage III/IV identification rather than a stage-and-age-independent prognostic marker. Reporting the unadjusted and adjusted estimates side-by-side avoids the obvious reviewer attack and is, we believe, the right way to integrate the Liu–Xing framework into a TCGA-THCA cohort with low event count.

The manuscript matches *npj Precision Oncology*'s scope on three fronts. First, the headline contribution is a **directly clinically deployable biomarker** with quantified outperformance over the standard-of-care molecular test (BRAF V600E), now corroborated by a Decision Curve Analysis showing dominance across the entire clinical threshold range and by 4-cohort meta-analytic transferability. Second, the underlying transcriptomic axis (DM1/DM2) is statistically distinct from BRAF/RAS mutation status (Spearman = 0.49 with the canonical dichotomy; 17 of 335 mutation-carrying tumours violate the canonical map) and maps to a hot/cold immune landscape with implications for both immunotherapy and TROP2-directed antibody-drug conjugate (ADC) stratification. Third, we propose evaluation as a correlative biomarker overlay in two ongoing thyroid TROP2-ADC trials (NCT06235216 SETHY, NCT07521670 STRAP), both of which currently accrue without molecular sub-stratification — translating the in-silico work into an immediately actionable trial-design recommendation.

This work represents a fully in-silico discovery from an independent Korean computational researcher (with part-time PhD affiliation at Seoul National University Graduate School of Convergence Science and Technology), in collaboration with [Yu Hyeong Won — full Korean/English name TBD by user] of [Affiliation TBD]. Korean cohort cross-validation via Seoul National University Hospital / Bundang Hospital is in active outreach for the revision round; the v4 4-cohort $I^2 = 0\%$ result strengthens but does not replace this independent prospective test.

We honestly report thirteen limitations: no wet-lab validation, no Korean cohort yet integrated, Cox HR for DM1/DM2 not significant for OS (consistent with TCGA-THCA's exceptional prognosis), 2/5 robust external transfer, scRNA cohort limited to 7 patients, PRISM cell-line drug screen underpowered at $n = 5$ vs $n = 5$, Thorsson cross-reference replaced by our composite, pan-cancer transfer is signature-overlap only, TERT calls from cBioPortal mirror (not BAM re-call), TERT subset only 6 events (with bootstrap and leave-one-out audit reported), the stage-adjustment caveat (univariate HR 6.31 $\rightarrow$ multivariate HR 1.88), the 4 meta-analysis cohorts overlap with axis discovery via GSE76039, and DCA dominance is a population-level result with calibration as the next individual-decision step. The 8-gene panel ΔAUC outperformance, now triply corroborated by meta-analysis + DCA + subgroup forest, is the central submission-defensible contribution.

Suggested reviewers (no conflicts of interest declared): - **Yuri Nikiforov, MD, PhD** — University of Pittsburgh — thyroid genomic classification. - **James A. Fagin, MD** — Memorial Sloan Kettering Cancer Center — BRAF biology and PTC molecular subtyping. - **Ricardo R. Lima, PhD** — PUC-RJ — thyroid sub-classification and Latin American cohorts. - **Mingzhao Xing, MD, PhD** — Johns Hopkins / SUSTech — Liu–Xing four-genotype prognostic framework foundational to our R8 integration.

All raw outputs, pipeline scripts, and reproducibility archives are available in the submission package (`anonymous_code.zip` for double-blind review). The interactive calculator and BOOST-sprint validation tables are at `reports/v17_boost/` and `results/v17_boost/`. The authors declare no competing interests. This work has not been submitted elsewhere.

Sincerely,

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v6 REAL FIX update (2026-04-27)

We additionally completed a critical leak-free re-validation of the 8-gene panel against the original DM1/DM2 cluster, which had been derived from the 67-gene TIERA67 panel that includes the 8 RAI genes — raising the possibility that the original CV AUC 0.954 reflected autocorrelation rather than genuine prediction. To address this proactively (before reviewer raises it), we re-defined the differentiation cluster four times, each time excluding the 8-gene panel: TIERA67 minus 8 (47 genes), 30-gene MAPK+immune+EMT (zero overlap with 8-gene panel), 5-marker pure-immune cluster, and BRS71-minus-8. The 8-gene panel achieves leak-free CV AUC = **0.962 (R1-A, 95% CI 0.940–0.979), 0.925 (R1-B zero-overlap), 0.957 (R1-D BRS framework)**, with ΔAUC over BRAF V600E sustained at +0.111 to +0.130 across all variants. The R1-B zero-overlap result (cluster definition shares zero genes with the 8-gene panel) is the strongest evidence that the panel captures genuine biological generalization, not autocorrelation. External validation on GSE76039 (histology ground truth) yields AUC = 0.935 (95% CI 0.824–1.000) under the leak-free model. We now frame the 8-gene panel honestly in v6 as a parsimonious classifier of the differentiation continuum, rather than as predicting an "independent" cluster.